

CEGM1000

Modelling, Uncertainty and Data for Engineers

Week 1.3

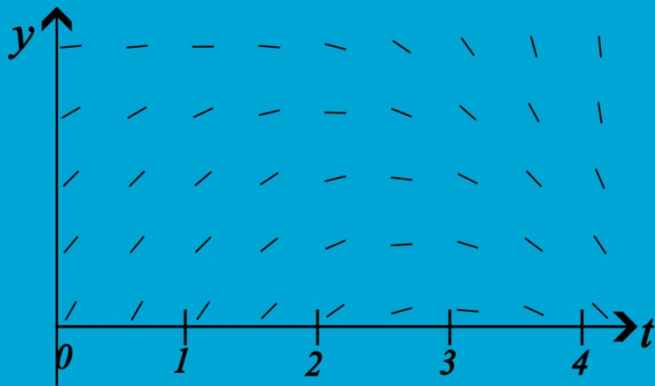
Numerical modelling

Part 2

Ronald Brinkgreve, Anna Störiko

Based on a previous version from Jaime Arriaga and
the rest of the MUDE team





Learning objectives

At the end of this lecture, you should be able to

- Discuss the characteristics of Explicit and Implicit Numerical schemes
- Schematize numerical solutions of ODEs
- Solve initial and boundary value problems numerically

Contents

- Explicit and Implicit numerical schemes:
single step
 - Initial Value Problems
- Multiple-step and multi-stage schemes
 - Initial Value Problems
- Second order ODEs
 - Boundary Value Problems

Back to the Falling Head test (initial value problem):

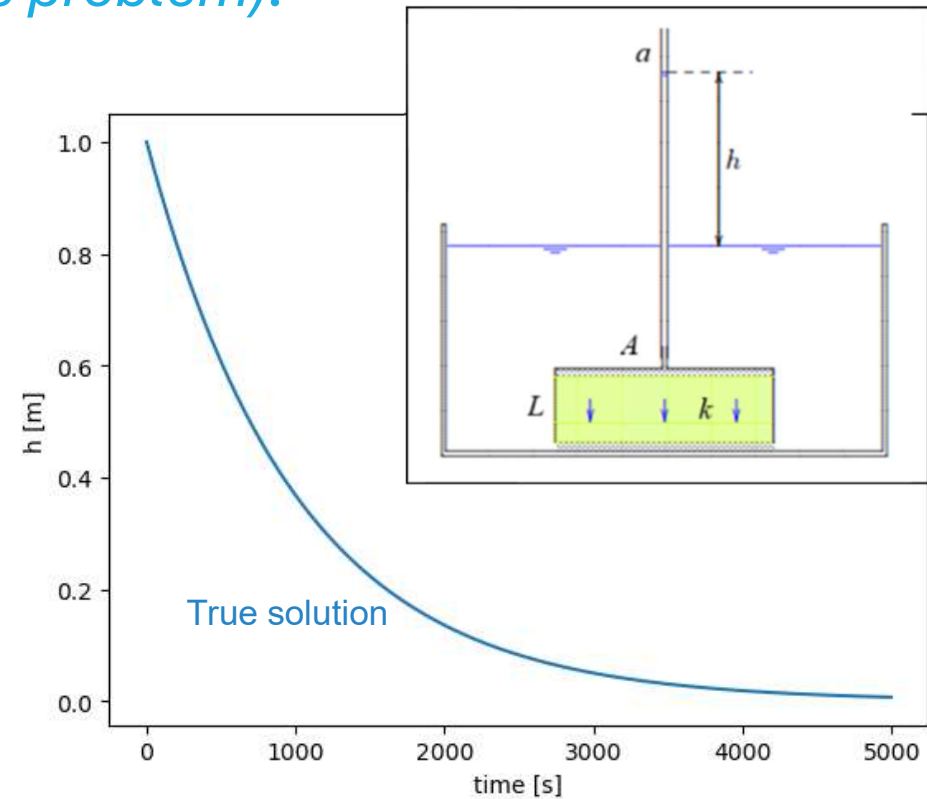
ODE:
$$\frac{dh}{dt} = -\frac{kA}{aL}h$$

Initial value: at $t = t_0$: $h = h_0$

Exact solution:
$$h(t) = h_0 e^{-\frac{kA}{aL}t}$$

k	10^{-6}	m/s
A	0.1	m ²
a	0.001	m ²
L	0.1	m
h_0	1.0	m

Solution depends on initial value (h_0) !



Purpose of the test on a soil sample:
To measure hydraulic conductivity k

Stability

ODE: $\frac{dh}{dt} = -\alpha h$ (in the example: $\alpha = \frac{kA}{aL}$)

Exact solution: $h(t) = h_0 e^{-\alpha t}$

Explicit Euler: $h_{i+1} = h_i - \alpha \Delta t h_i = h_i(1 - \alpha \Delta t)$

Step 1: $h_1 = h_0(1 - \alpha \Delta t)$

Step 2: $h_2 = h_1(1 - \alpha \Delta t) = h_0(1 - \alpha \Delta t)^2$

Step 3: $h_3 = h_2(1 - \alpha \Delta t) = h_0(1 - \alpha \Delta t)^3$

Step n : $h_n = h_0 (1 - \alpha \Delta t)^n$

Stability: $|1 - \alpha \Delta t| < 1 \leftrightarrow 0 < \alpha \Delta t < 2 \rightarrow \Delta t < 2/\alpha$

Stability

ODE: $\frac{dh}{dt} = -\alpha h$ (in the example: $\alpha = \frac{kA}{aL}$)

Exact solution: $h(t) = h_0 e^{-\alpha t}$

Stability criterion for **Explicit Euler**: $\Delta t < \frac{2}{\alpha}$

Note that the solution may already become highly inaccurate before the limit!

Implicit Euler is unconditionally stable; this does NOT mean it is more accurate!

The Falling Head test

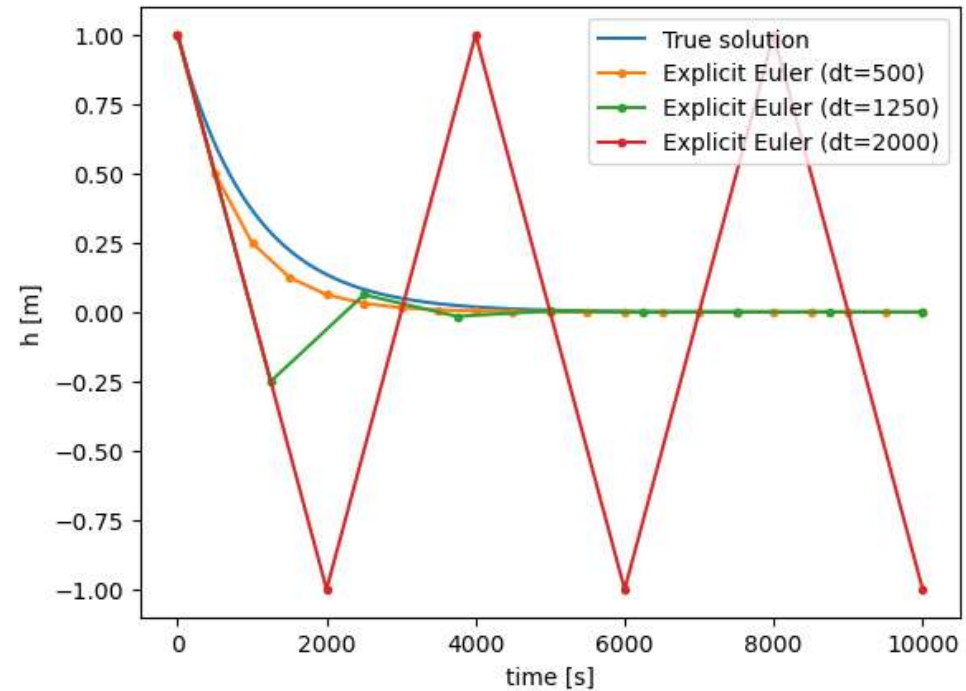
ODE: $\frac{dh}{dt} = -\frac{kA}{aL}h$

Initial value: at $t = t_0$: $h = h_0$

Explicit Euler: $\frac{h(t_{i+1}) - h(t_i)}{\Delta t} = -\frac{kA}{aL}h(t_i)$

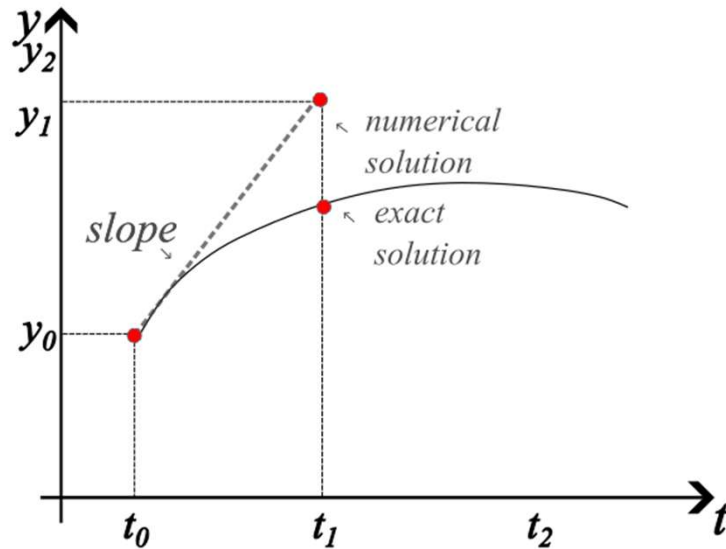
$$\Rightarrow h(t_{i+1}) = h(t_i) - \Delta t \frac{kA}{aL} h(t_i)$$
$$= \left(1 - \Delta t \frac{kA}{aL}\right) h(t_i)$$

Stability: $\Delta t < 2 \frac{aL}{kA} = 2000 \text{ s}$



```
t_n = np.linspace(0, maxtime, n_steps + 1)
h_EF = []
h_EF.append(h0)
for i in range(n_steps):
    h_new = (1.0 - dt * k * A / (a * L)) * h_EF[-1]
    h_EF.append(h_new)
```

Explicit Euler (Euler Forward)



$$t_{i+1} = t_i + \Delta t$$

$$y_{i+1} = y_i + \Delta t * \text{slope}_i$$

$$y_1 = y_0 + \Delta t * \text{slope}_0$$

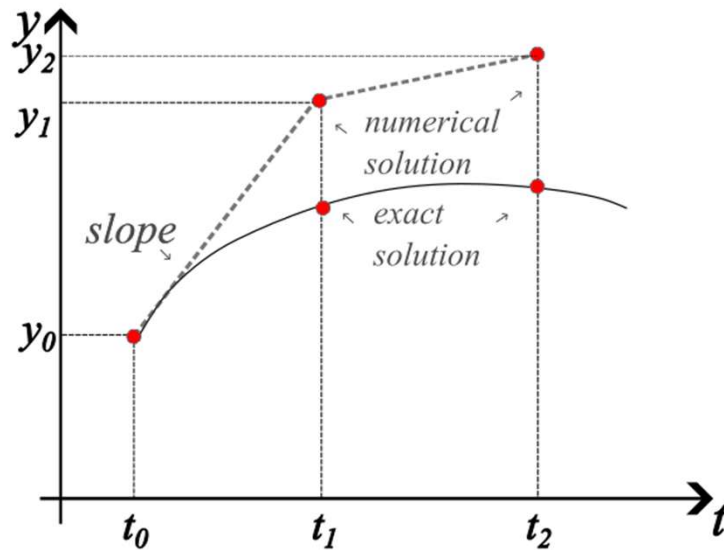
The slope is computed using the Forward Difference formula, first-order accurate.

$$\text{Truncation error} = y_1^{TSE} - y_1^{FE}$$

$$= y_0 + \Delta t y'_0 + \frac{\Delta t^2}{2!} y''_0 + \dots - (y_0 + \Delta t y'_0)$$

$$= \frac{\Delta t^2}{2!} y''_0 + \dots \approx O(\Delta t^2)$$

Explicit Euler (Euler Forward)



$$t_{i+1} = t_i + \Delta t$$

$$y_{i+1} = y_i + \Delta t * \text{slope}_i$$

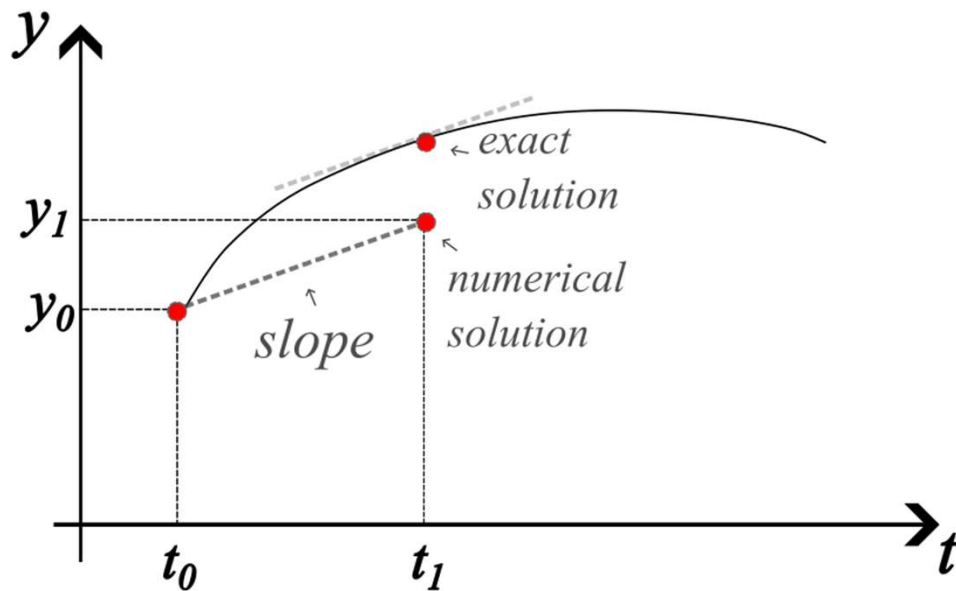
$$y_2 = y_1 + \Delta t * \text{slope}_1$$

$$\text{Total truncation error} = \sum_{i=0}^{n-1} y_{i+1}^{TSE} - y_{i+1}^{FE}$$

$$= \sum_{i=0}^{n-1} \frac{\Delta t^2}{2!} y''_i + \dots \approx \frac{\Delta t^2}{2!} \frac{b-a}{\Delta t} \overline{y''}$$

$$\approx \frac{\Delta t}{2!} (b-a) \overline{y''} \approx O(\Delta t)$$

Implicit Euler (Euler Backward)



$$t_{i+1} = t_i + \Delta t$$

$$y_{i+1} = y_i + \Delta t * slope_{i+1}$$

$$y_1 = y_0 + \Delta t * slope_1$$

The slope is computed using the Backward Difference formula, first-order accurate.

Total truncation error $\approx O(\Delta t)$

Back to our example: The Falling Head test

ODE:
$$\frac{dh}{dt} = -\frac{kA}{aL}h$$

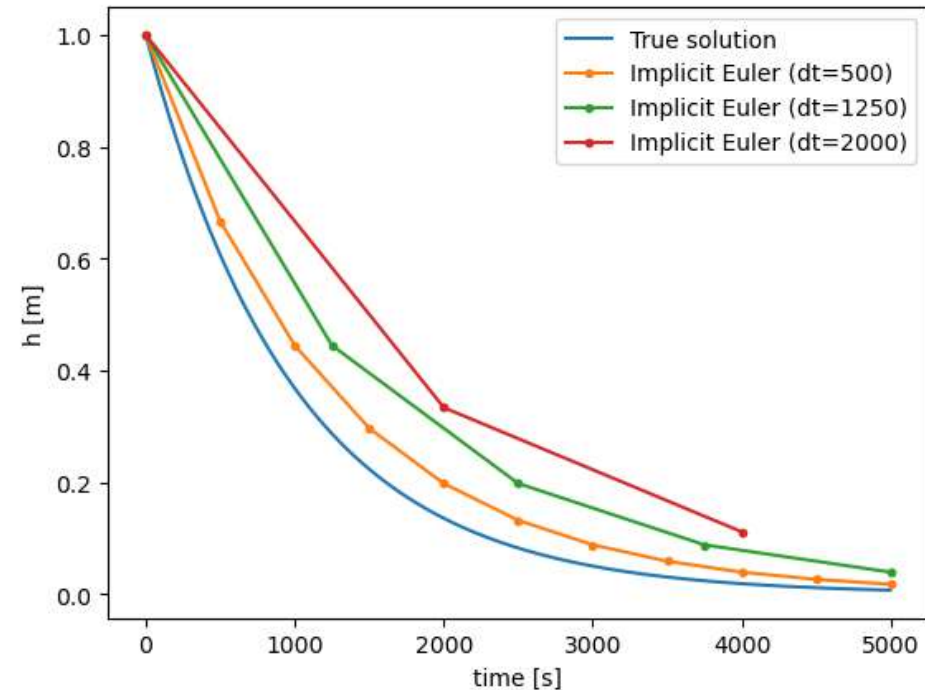
Initial value: $\text{at } t = t_0: h = h_0$

Implicit Euler:
$$\frac{h(t_{i+1}) - h(t_i)}{\Delta t} = -\frac{kA}{aL}h(t_{i+1})$$

$$\Rightarrow h(t_{i+1}) \left(1 + \Delta t \frac{kA}{aL} \right) = h(t_i)$$

$$h(t_{i+1}) = h(t_i) / \left(1 + \Delta t \frac{kA}{aL} \right)$$

Stability: Not an issue (unconditionally stable)
This does not mean it is more accurate!



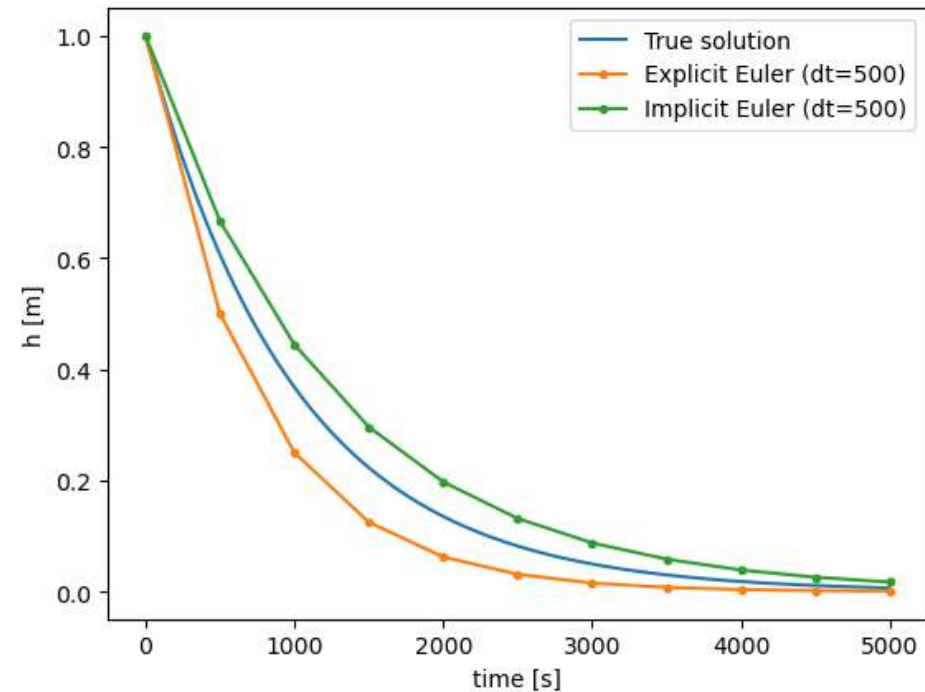
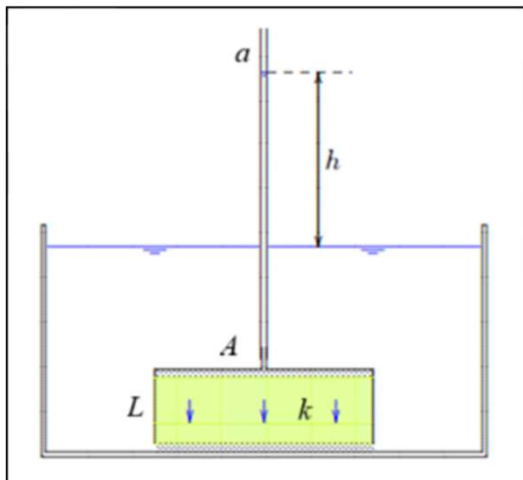
```
h_EB = []
h_EB.append(h0)
for i in range(n_steps):
    h_new = h_EB[-1] / (1.0 + dt * k * A / (a * L))
    h_EB.append(h_new)
```

Back to our example: The Falling Head test

ODE:
$$\frac{dh}{dt} = -\frac{kA}{aL}h$$

Initial value: at $t = t_0$: $h = h_0$

Comparison of solutions:



Explicit: 'Faster' than true solution

Implicit: 'Slower' than true solution

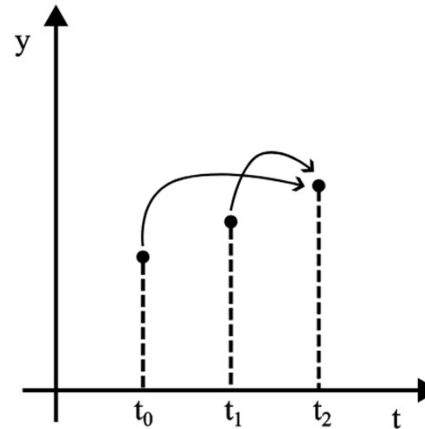
('faster' or 'slower' depends on the type of problem)

Multi-step methods

- Adams-Bashforth is an explicit method of higher accuracy
- AB is not self starting!
- Adams-Moulton is an implicit method of higher accuracy
- AM is self starting but requires iterations

Adams-Bashforth
second order accurate

$$y_{i+1} = y_i + \frac{\Delta t}{2} (3y'_i - y'_{i-1})$$



Not enough info at t_1

Multi-step means:
To calculate a new point, it uses
more than one previous point

Multi-stage methods

The main error of simple single step methods is assuming that the slope does not change between i and $i + 1$

- Modified Euler

$$y_{i+1} = y_i + \Delta t (y'_i + y'_{i+1*})/2$$

- Midpoint

$$y_{i+1} = y_i + \Delta t y'_{i+\frac{1}{2}}$$

- Heun's method
(= 2-stage RK)

$$y_{i+1} = y_i + \Delta t (k_1 + k_2) / 2$$

$$k_1 = y'_i$$

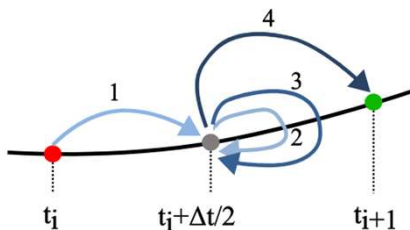
$$k_2 = y'_{i+1*} \text{ with } y_{i+1*} = y_i + \Delta t k_1$$

- 4-stage Runge-Kutta

$$y_{i+1} = y_i + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4) + \mathcal{O}(\Delta t^5)$$

$$\begin{cases} k_1 = y'_i \\ k_2 = y'_{i+1/2} \text{ with } y_{i+1/2} = y_i + \frac{\Delta t}{2} k_1 \\ k_3 = y'_{i+1/2} \text{ with } y_{i+1/2} = y_i + \frac{\Delta t}{2} k_2 \\ k_4 = y'_{i+1} \text{ with } y_{i+1} = y_i + \Delta t k_3 \end{cases}$$

Multi-stage means:
To calculate a new point, it
uses intermediate estimates
of the new point



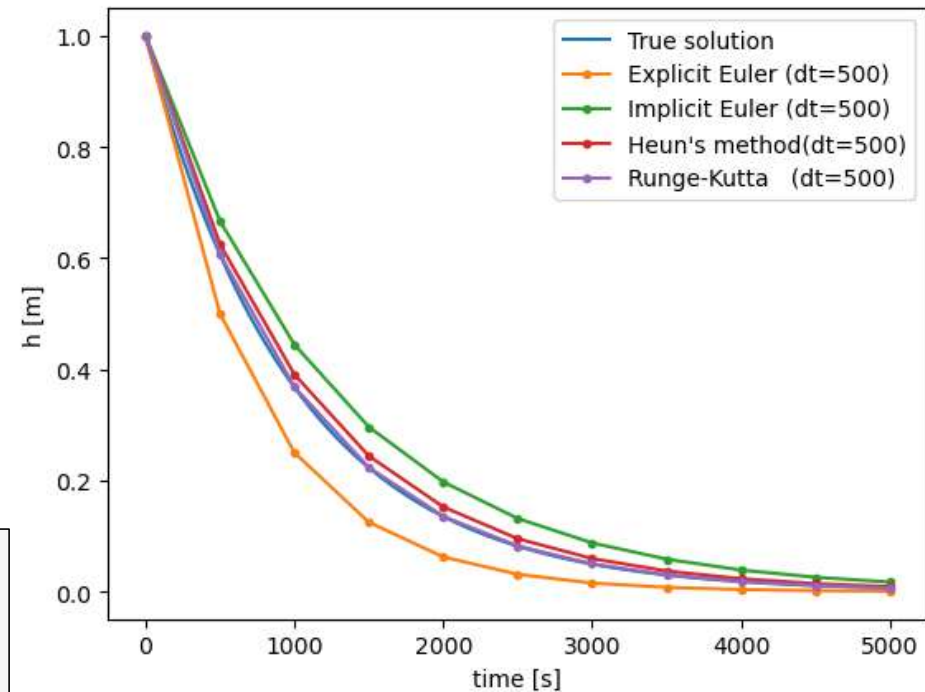
Back to our example: The Falling Head test

ODE:
$$\frac{dh}{dt} = -\frac{kA}{aL}h$$

Initial value: at $t = t_0$: $h = h_0$

Comparison of solutions:

```
h_RK4 = []
h_RK4.append(h0)
for i in range(n_steps):
    fac = -k * A / (a * L)
    k1 = fac * h_RK4[-1]
    k2 = fac * (h_RK4[-1] + dt / 2 * k1)
    k3 = fac * (h_RK4[-1] + dt / 2 * k2)
    k4 = fac * (h_RK4[-1] + dt * k3)
    h_new = h_RK4[-1] + dt / 6 * (k1 + 2 * k2 + 2 * k3 + k4)
    h_RK4.append(h_new)
```



Non-linear ODE

$$\frac{dp(t)}{dt} = -p^{3/2} + 5 \cdot p_{cst}(1 - e^{-t})$$

Euler forward (explicit):

$$p_{i+1} = p_i + \Delta t \left(-p_i^{3/2} + 5p_{cst} \cdot (1 - e^{-t_i}) \right)$$

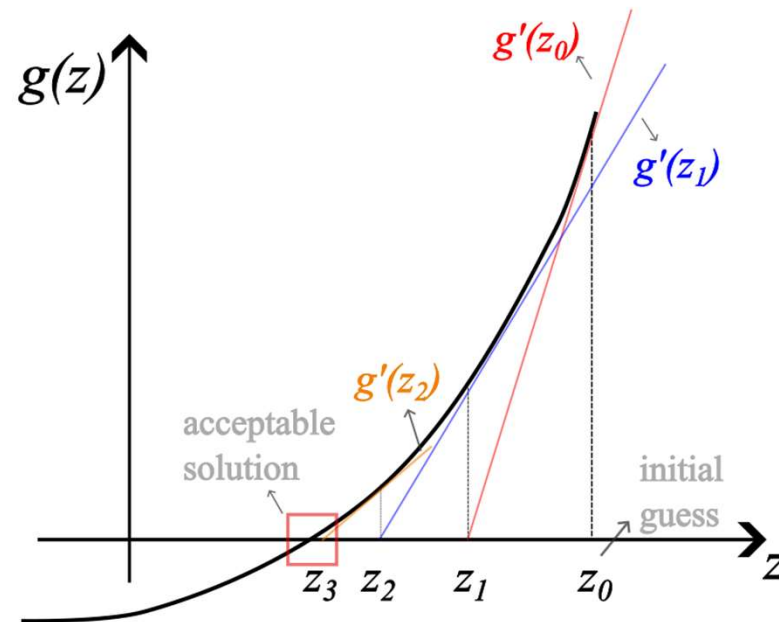
Can be solved directly

Euler backward (implicit):

$$p_{i+1} = p_i + \Delta t \left(-p_{i+1}^{3/2} + 5p_{cst} * (1 - e^{-t_{i+1}}) \right)$$

Can only be solved iteratively

Non-linear ODE: Newton-Raphson (iterative) method for Euler Backward (implicit)



Newton-Raphson updating:

$$z_{i+1} = z_i - \frac{g(z_i)}{g'(z_i)}$$

Example: Non-linear stress-strain relation in one-dimensional compression

Oedometer test: to measure soil compressibility (or stiffness)

Stress-strain relationship:

$$d\sigma = C \left(\frac{\sigma}{\sigma^{ref}} \right)^m d\varepsilon$$

└ stiffness at ref. stress

As ODE:

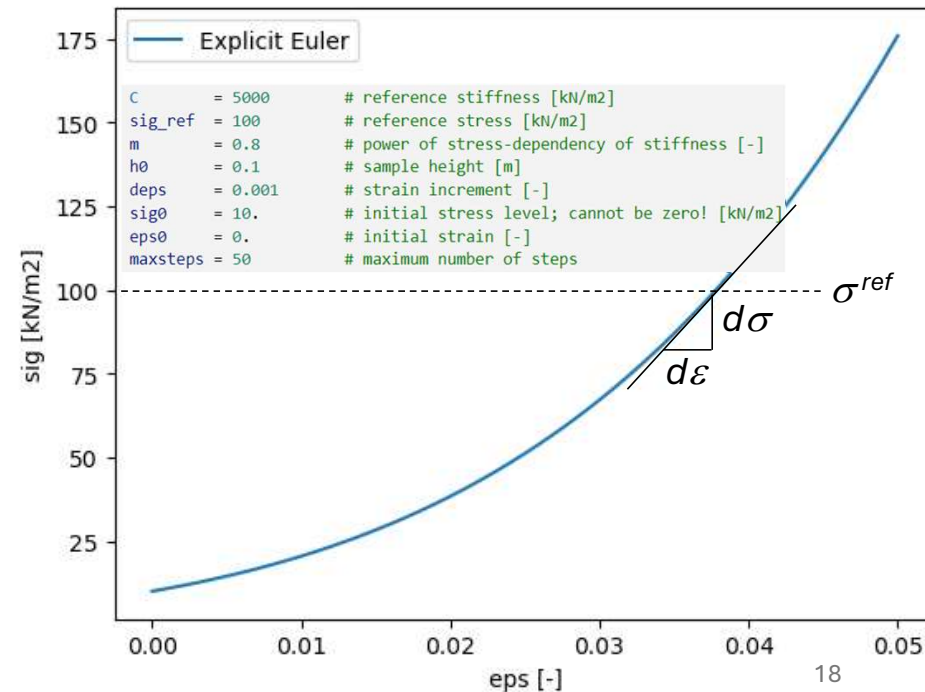
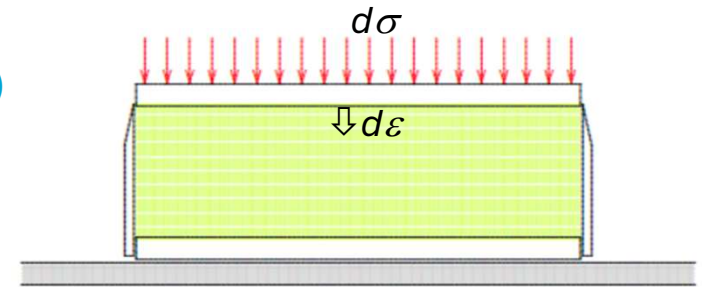
$$\frac{d\sigma}{d\varepsilon} = C \left(\frac{\sigma}{\sigma^{ref}} \right)^m = C^* \sigma^m$$

with $C^* = \frac{C}{(\sigma^{ref})^m}$

Explicit Euler:

$$\frac{\sigma_{i+1} - \sigma_i}{\Delta\varepsilon} = C^* \sigma_i^m$$

$$\sigma_{i+1} = \sigma_i + \Delta\varepsilon C^* \sigma_i^m$$

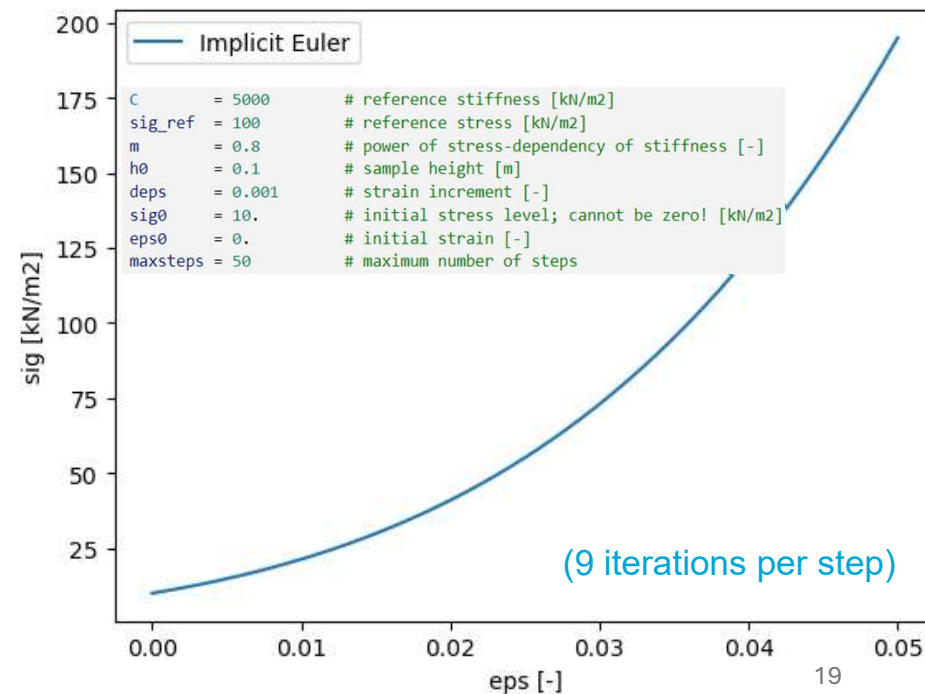
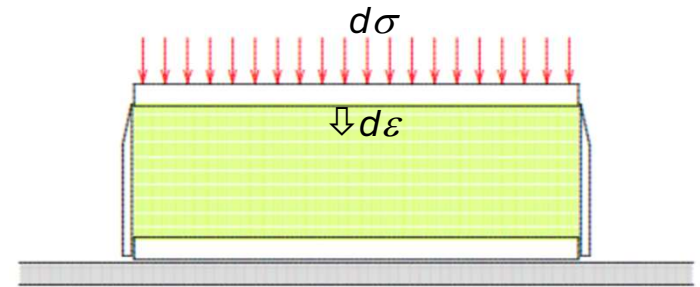


Example: Non-linear stress-strain relation in one-dimensional compression

ODE:
$$\frac{d\sigma}{d\varepsilon} = C \left(\frac{\sigma}{\sigma_{ref}} \right)^m = C^* \sigma^m$$

Implicit Euler:
$$\frac{\sigma_{i+1} - \sigma_i}{\Delta\varepsilon} = C^* (\sigma_{i+1})^m$$

Possible solution:
(solved iteratively)
$$\sigma_{i+1}^{j+1} = \sigma_i + \Delta\varepsilon C^* (\sigma_{i+1}^j)^m$$



Example: Non-linear stress-strain relation in one-dimensional compression

ODE:
$$\frac{d\sigma}{d\varepsilon} = C \left(\frac{\sigma}{\sigma^{ref}} \right)^m = C^* \sigma^m$$

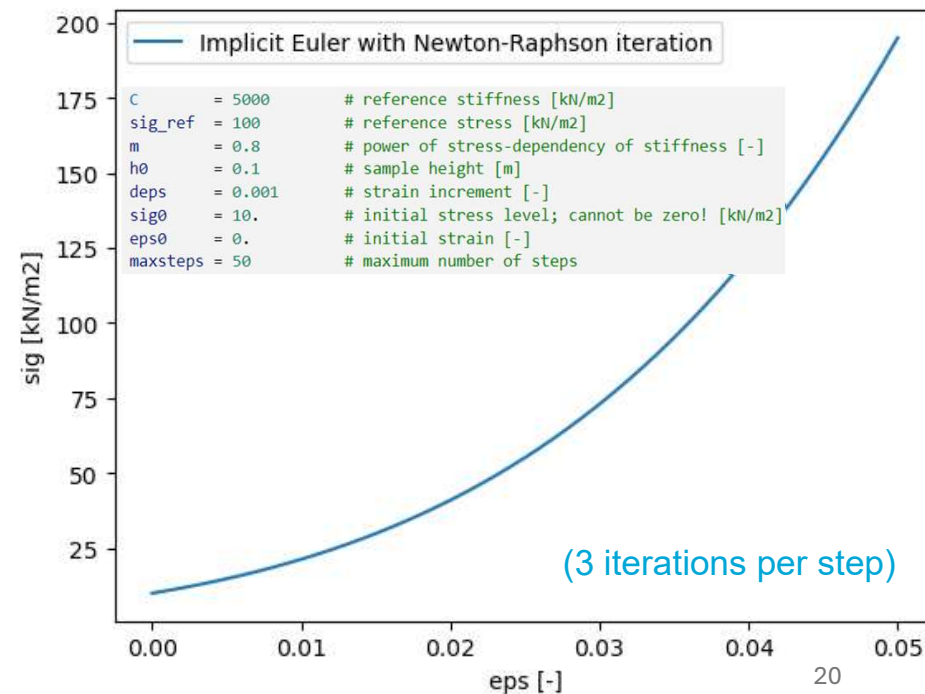
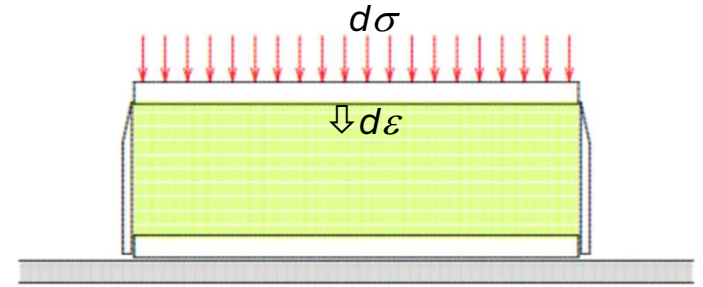
Best solution (least number of iterations):
Implicit Euler with Newton-Raphson iterations:

$$g(\sigma_{i+1}) = 0 \quad \text{with} \quad g(\sigma_{i+1}) = \sigma_{i+1} - \Delta\varepsilon C^* (\sigma_{i+1})^m - \sigma_i$$

$$\sigma_{i+1}^{j+1} = \sigma_{i+1}^j - \frac{g(\sigma_{i+1}^j)}{g'(\sigma_{i+1}^j)}$$

$$g(\sigma_{i+1}^j) = \sigma_{i+1}^j - \Delta\varepsilon C^* (\sigma_{i+1}^j)^m - \sigma_i$$

$$g'(\sigma_{i+1}^j) = \frac{dg}{d\sigma_{i+1}} = 1 - m\Delta\varepsilon C^* (\sigma_{i+1}^j)^{m-1}$$



Example: Non-linear stress-strain relation in one-dimensional compression

ODE:
$$\frac{d\sigma}{d\varepsilon} = C \left(\frac{\sigma}{\sigma_{ref}} \right)^m = C^* \sigma^m$$

Comparison of solutions:

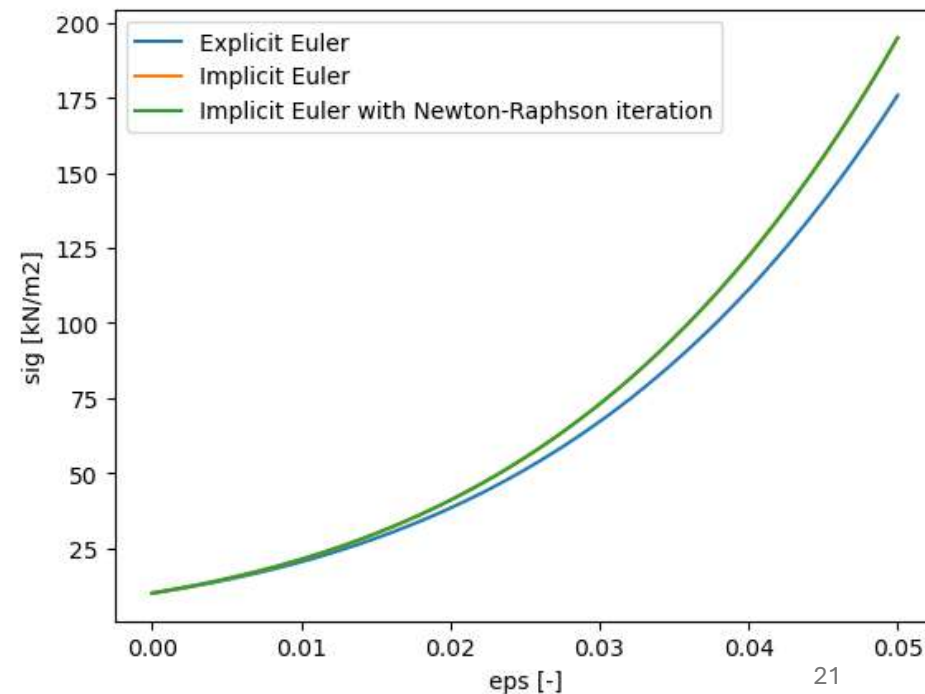
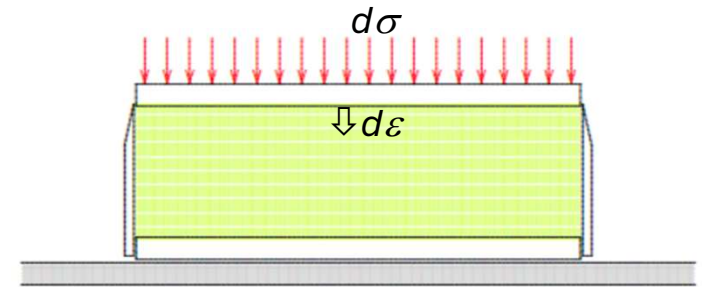
```

C      = 5000      # reference stiffness [kN/m2]
sig_ref = 100      # reference stress [kN/m2]
m      = 0.8       # power of stress-dependency of stiffness [-]
h0     = 0.1       # sample height [m]
deps   = 0.001     # strain increment [-]
sig0    = 10.      # initial stress level; cannot be zero! [kN/m2]
eps0    = 0.        # initial strain [-]
maxsteps = 50      # maximum number of steps
    
```

Final sample height:

$$h_{end} = h_0 (1 - \epsilon) = 0.095 \text{ m}$$

Explicit: Solution is 'softer'
 Implicit: Solution is 'stiffer'
 ('true' solution is in between)



Let's take a break..

Please, take a seat...



Second order ODE

$$\frac{d^2 y}{dt^2} = \frac{dy}{dt} - y + \cos t = 0 \quad \rightarrow \text{IVP}$$

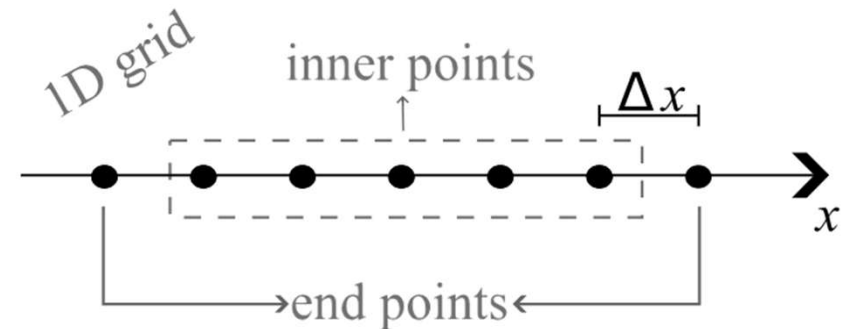
(For example: Cyclic motion of a mass-damping system with external cyclic excitation)

1D steady-state heat distribution along a bar:

$$\frac{d^2 T}{dx^2} - \alpha_1(T - T_s) = 0 \quad \rightarrow \text{BVP}$$

Heat convection to the outside medium

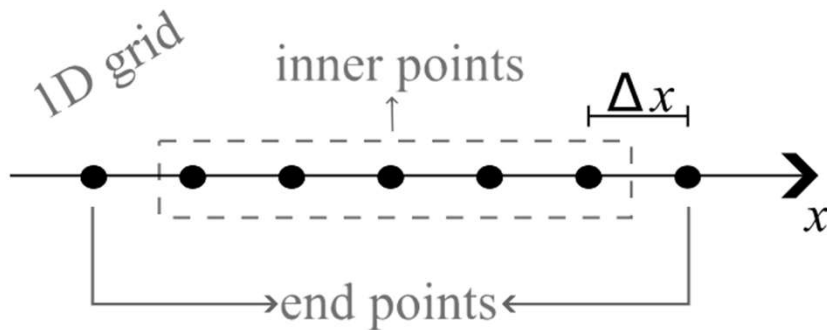
Heat conduction within the bar



Second order ODE as BVP: Boundary Conditions

$$\frac{d^2 T}{dx^2} - \alpha_1(T - T_s) = 0$$

$$\frac{d^2 y}{dx^2} = g\left(x, y, \frac{dy}{dx}\right)$$



- **Dirichlet**

$$y(x = a) = Y_a \text{ and } y(x = b) = Y_b$$

- **Neumann**

$$\left. \frac{dy}{dx} \right|_{x=a} = D_a \text{ and } \left. \frac{dy}{dx} \right|_{x=b} = D_b$$

- **Mixed**

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Question slide

Give an example of a Neumann boundary condition for a deformation or a flow problem



##/##

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Results slide

Give an example of a Neumann boundary condition for a deformation or a flow problem

RESULTS SLIDE

Exercise – Temperature distribution across a thin plate

Consider the following differential equation:

$$\frac{d^2T}{dx^2} = \alpha(T - T_s) \quad \text{or} \quad \frac{d^2T}{dx^2} - \alpha(T - T_s) = 0$$

in the domain $x \in (0, 0.1)$

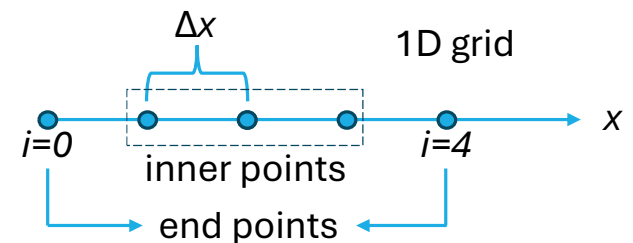
with the conditions: $T(0) = 473[K]$, $T(0.1) = 293[K]$

What type of boundary conditions are these?

Central Difference scheme:

$$f''(x_0) \approx \frac{f(x_0 - \Delta x) - 2f(x_0) + f(x_0 + \Delta x)}{(\Delta x)^2}$$

$$\text{or} \quad f''(x_i) \approx \frac{f(x_{i-1}) - 2f(x_i) + f(x_{i+1}))}{(\Delta x)^2}$$



Elaborate the general numerical equation in T_{i-1} , T_i , T_{i+1} according to the Central Difference scheme.

Elaborate the scheme further using 3 inner grid points and add the boundary conditions.

(i.e. consider $i = 1, 2, 3$ in the equation and apply the boundary conditions at the end points $i = 0, 4$)

Formulate the result as a matrix-vector system.

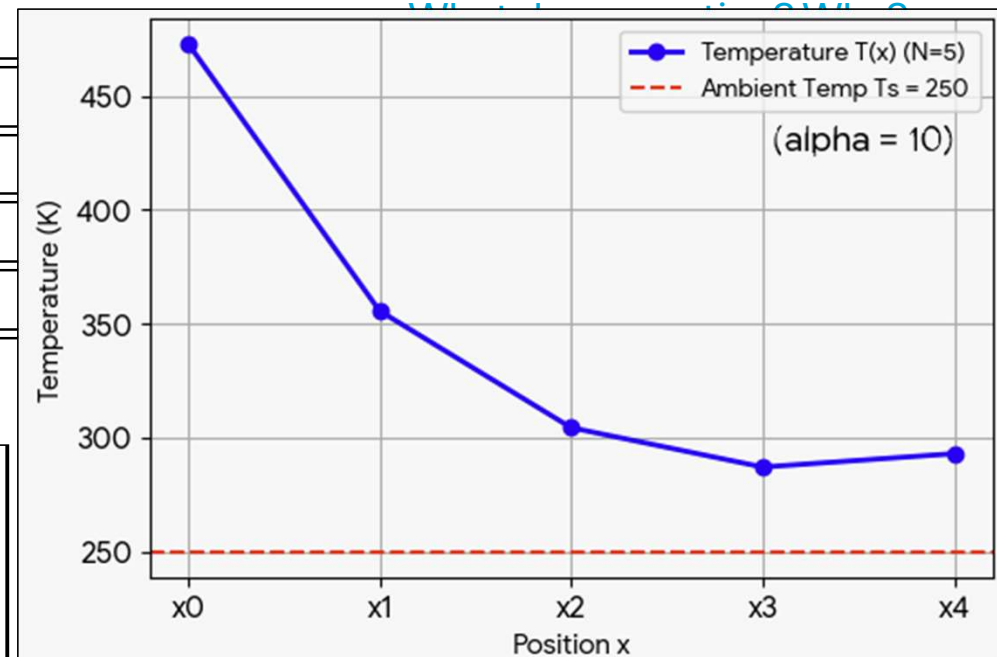
Exercise – Temperature distribution across a thin plate

$$\frac{T_{i-1} - 2T_i + T_{i+1}}{\Delta x^2} - \alpha(T_i - T_s) = 0 \quad \Leftrightarrow \quad T_{i-1} - 2T_i + T_{i+1} - \alpha\Delta x^2(T_i - T_s) = 0$$

$$T_{i-1} - (2 + \alpha\Delta x^2)T_i + T_{i+1} = -\alpha\Delta x^2 T_s$$

$$\begin{aligned} & \boxed{T_0} - (2 + \alpha\Delta x^2)\boxed{T_1} + \boxed{T_2} \\ & \boxed{T_1} - (2 + \alpha\Delta x^2)\boxed{T_2} + \boxed{T_3} \\ & \boxed{T_2} - (2 + \alpha\Delta x^2)\boxed{T_3} + \boxed{T_4} \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & -(2 + \alpha\Delta x^2) & 1 & 0 & 0 \\ 0 & 1 & -(2 + \alpha\Delta x^2) & 1 & 0 \\ 0 & 0 & 1 & -(2 + \alpha\Delta x^2) & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} T_0 \\ T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix}$$



Example: Euler-Bernoulli beam deflection

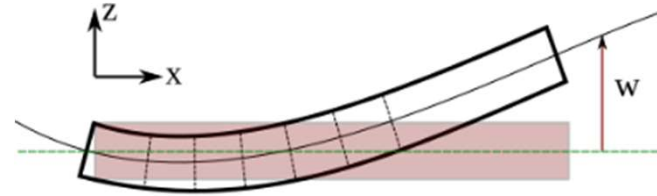
$$\frac{d^2 M}{dx^2} = -q \quad \text{and} \quad M = -EI \frac{d^2 w}{dx^2}$$

M = bending moment

w = beam deflection (lateral displacement)

EI = flexural rigidity (bending stiffness)

q = distributed load



Central difference scheme for both equations:

$$\frac{M_{i-1} - 2M_i + M_{i+1}}{\Delta x^2} = -q_i \quad \text{or} \quad M_{i-1} - 2M_i + M_{i+1} = -\Delta x^2 q_i \quad \text{or} \quad -M_{i-1} + 2M_i - M_{i+1} = \Delta x^2 q_i$$

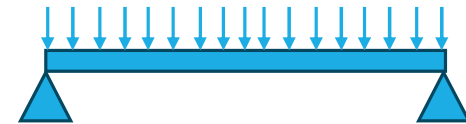
$$-\frac{M_i}{EI} = \frac{w_{i-1} - 2w_i + w_{i+1}}{\Delta x^2} \quad \text{or} \quad w_{i-1} - 2w_i + w_{i+1} = -\frac{\Delta x^2}{EI} M_i \quad \text{or} \quad -w_{i-1} + 2w_i - w_{i+1} = \frac{\Delta x^2}{EI} M_i$$

Example: Euler-Bernoulli beam deflection

Consider a simply supported beam at both ends with the following boundary conditions:

at $x = 0$: $w = 0$ and $M = 0$

at $x = L$: $w = 0$ and $M = 0$



First, solve the bending moments:

$$-M_{i-1} + 2M_i - M_{i+1} = \Delta x^2 q_i$$

This gives a system of equations:

$$\mathbf{A} \mathbf{M} = \mathbf{f} \quad \Rightarrow \quad \mathbf{M} = \mathbf{A}^{-1} \mathbf{f}$$

(in Python: $\mathbf{M} = \text{numpy.linalg.solve}(\mathbf{A}, \mathbf{f})$)

$$\begin{bmatrix}
 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\
 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\
 & & \ddots & \ddots & \ddots & & & \\
 & & & -1 & 2 & -1 & 0 & 0 \\
 & 0 & & 0 & -1 & 2 & -1 & 0 \\
 & & & 0 & 0 & 0 & 1 & 0
 \end{bmatrix}
 \begin{bmatrix} M_1 \\ M_2 \\ M_3 \\ M_4 \\ M_5 \\ M_6 \\ M_7 \\ M_8 \end{bmatrix} = \begin{bmatrix} 0 \\ \Delta x^2 q_1 \\ \Delta x^2 q_2 \\ \Delta x^2 q_3 \\ \Delta x^2 q_4 \\ \Delta x^2 q_5 \\ \Delta x^2 q_6 \\ 0 \end{bmatrix}$$

Diagram illustrating the 1D grid and boundary conditions. The grid consists of points along the x-axis. The first and last points are labeled "end points" and are connected by a bracket. The points between them are labeled "inner points". The distance between adjacent points is Δx . The boundary conditions are indicated by blue arrows pointing to the first and last points of the grid, corresponding to the first and last rows of the matrix equation.

Example: Euler-Bernoulli beam deflection

Consider a simply supported beam at both ends with the following boundary conditions:

at $x = 0$: $w = 0$ and $M = 0$

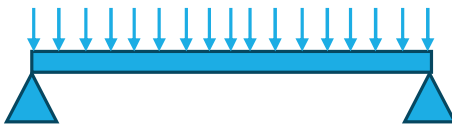
at $x = L$: $w = 0$ and $M = 0$

Then, solve the beam deflections:

$$-w_{i-1} + 2w_i - w_{i+1} = \frac{\Delta x^2}{EI} M_i$$

This gives a system of equations:

$$\mathbf{A} \mathbf{w} = \mathbf{f} \Rightarrow \mathbf{w} = \mathbf{A}^{-1} \mathbf{f}$$



(in Python: `w = numpy.linalg.solve(A,f)`)

$$\begin{bmatrix}
 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\
 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\
 & & \ddots & \ddots & \ddots & & & \\
 & & & -1 & 2 & -1 & 0 & 0 \\
 0 & & & 0 & -1 & 2 & -1 & 0 \\
 & & & 0 & 0 & 0 & 1 & 0
 \end{bmatrix}
 \begin{bmatrix}
 w_1 \\
 w_2 \\
 w_3 \\
 \vdots \\
 w_i \\
 \vdots \\
 w_{n-1} \\
 w_n
 \end{bmatrix}
 =
 \begin{bmatrix}
 0 \\
 \Delta x^2 M_1 / EI \\
 \Delta x^2 M_2 / EI \\
 \vdots \\
 \Delta x^2 M_i / EI \\
 \vdots \\
 \Delta x^2 M_{n-1} / EI \\
 0
 \end{bmatrix}$$

1D grid

inner points

Δx

end points

Boundary conditions

x

Example: Euler-Bernoulli beam deflection

Alternatively, both differential equations can be solved simultaneously:

$$-M_{i-1} + 2M_i - M_{i+1} = \Delta x^2 q_i$$

$$-w_{i-1} + 2w_i - w_{i+1} = \frac{\Delta x^2}{EI} M_i$$

$\Downarrow$

$$-\frac{\Delta x^2}{EI} M_i - w_{i-1} + 2w_i - w_{i+1} = 0$$

(RHS term is now integrated in the matrix)

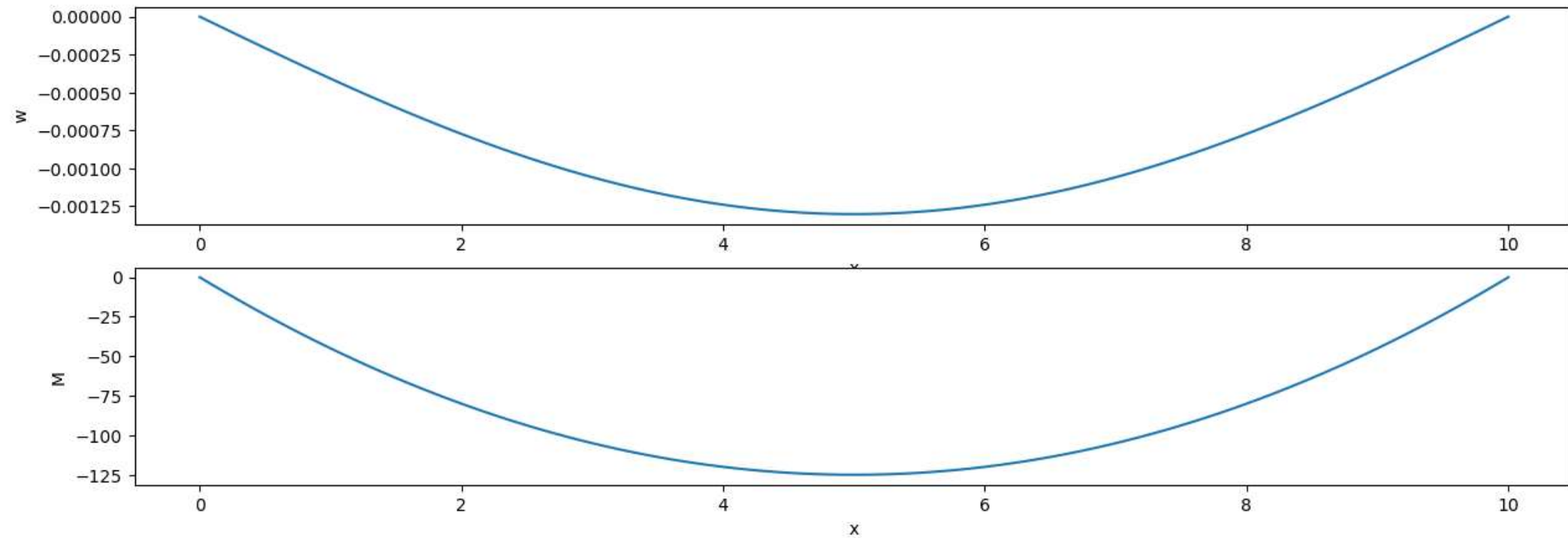
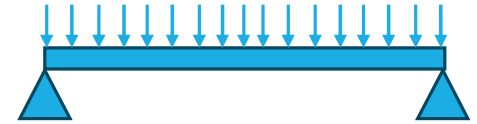
Example: Euler-Bernoulli beam deflection

Alternatively, both differential equations can be solved simultaneously:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ & & \ddots & \ddots \\ & & & -1 & 2 & -1 & 0 \\ & & & 0 & -1 & 2 & -1 \\ & & & 0 & 0 & 0 & 1 \\ 0 & -\Delta x^2/EI & & & & & & 1 \\ & & -\Delta x^2/EI & & & & & -1 \\ & & & \ddots & & & & \\ & & & & -\Delta x^2/EI & & & \\ & & & & & -\Delta x^2/EI & & \\ & & & & & & 0 & 1 \end{bmatrix} \begin{bmatrix} M_i \\ \\ \\ w_i \end{bmatrix} = \begin{bmatrix} 0 \\ -\Delta x^2 q_i \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Example: Euler-Bernoulli beam deflection

Results for $L=10\text{m}$, $EI=10^6\text{ kNm}^2$, $q=-10\text{ kN/m}^2$, $\Delta x=0.1$



Summary – Please, keep your attention! More info will follow!

- Differential equations:
 - Initial Value Problems
 - Boundary Value Problems
 - Example: Falling head test
- Numerical solutions:
 - Explicit Euler Stability criterion!
 - Implicit Euler Unconditionally stable!
- Non-linear ODE using implicit scheme: Iterative solution (Newton-Raphson):
- Multiple-step and multi-stage schemes
 - Heun
 - Runge-Kutta
- Second order ODEs
 - Boundary Value Problems
 - Example: Bending beam

What you can expect further this week...

- Programming Assignment 1.3:
 - VS Code IntelliSense and coding collaboration using VS share
 - Programming preparations for workshop and group assignments: matrix and vector manipulations
 - Fundamental programming concept: modules and numpy
- Wednesday 10:45-12:30: Workshop 1.3:
 - Solving boundary value problem: Bending beam
 - Solving two coupled ODEs in one system
 - Solving non-linear ODEs: Particle tracking
- Friday 8:45-12:30: Group Assignment 1.3:
 - Solving non-linear ODE for contaminant degradation

Interested in **Geotechnical Engineering**?

Join us in a series of events to **meet with experts** in the field:

Monday 14 September, 16:00-17:00, CEG Hall C: **GeoTalks**

- **Anne-Catherine Dieudonné**
Towards deep geological disposal: the geo-mechanical challenge of tunnelling in poorly indurated clay
- **Ken Gavin**
Foundation challenge in offshore renewables projects
- **Tristan Quinten**
Centrifuge Modelling for Offshore Geotechnics: Bridging the Gap Between the Laboratory and Offshore Renewable Energy

Monday 14 September, 17:00-19:00, Exhibition Room: **GeoDrinks**

Tuesday 15 September 12:45-13:30, CEG Hall D: **Track info**

- **Introduction to the field of Geotechnical Engineering and the Master Track**